SPACEX FALCON 9

First-Stage Landing Prediction

IBM Applied Data Science Capstone

Prepared using IBM course-prepared SpaceX launch data

Project Objective

Problem definition and analytical approach

Objective: predict whether a Falcon 9 first stage lands successfully (Class 1) or unsuccessfully (Class 0).

Why it matters: booster recovery and reuse are central to reducing launch costs.

Workflow: inspect and prepare data → explore patterns → perform SQL analysis → visualize launch sites → train and compare classifiers.

Dataset source: IBM Skills Network course-prepared SpaceX datasets; this run uses 90 labelled launch records.

Dataset & Preparation

The model was trained on the course-prepared labelled dataset

90	21	80 / 20	3-fold
labelled records	encoded predictors	train–test split	GridSearchCV

Target variable: Class (0 = failure, 1 = success).

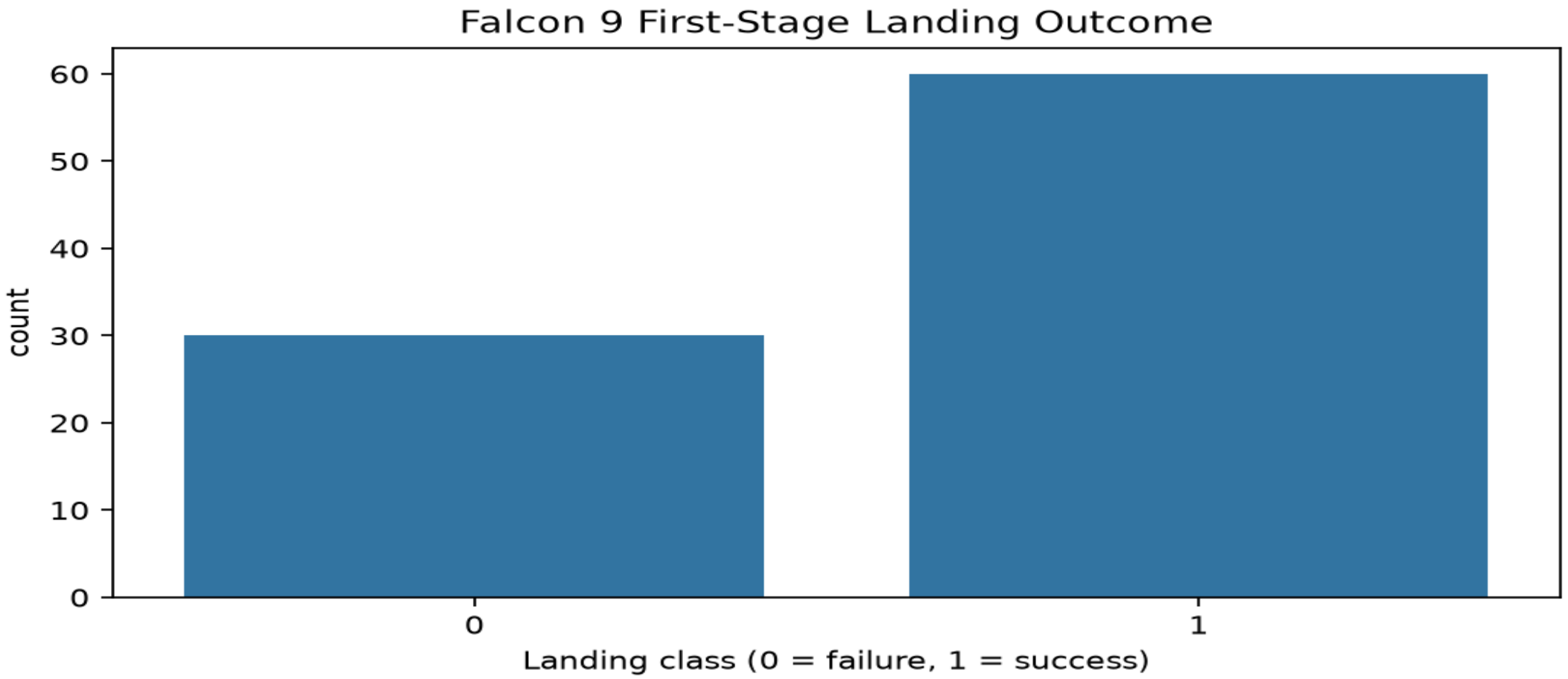
Rows without a target value were excluded; categorical variables were one-hot encoded.

Missing numeric values were imputed using the median; model predictors were prepared for classification.

Evaluation used a held-out test set with random_state = 42.

Exploratory Data Analysis — Landing Outcomes

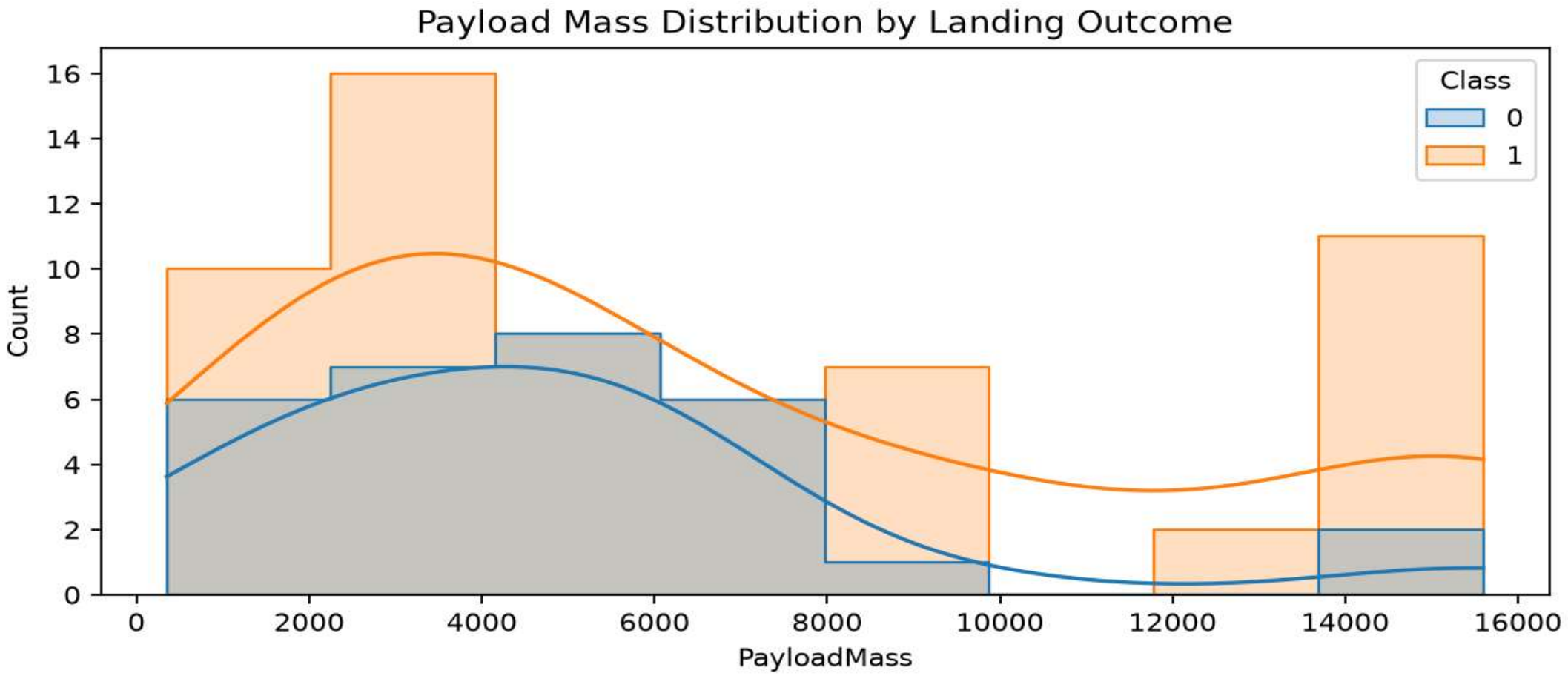
The labelled sample contains more successful than unsuccessful landings.



Observed counts: 30 unsuccessful outcomes (Class 0) and 60 successful outcomes (Class 1).

Exploratory Data Analysis — Payload Mass

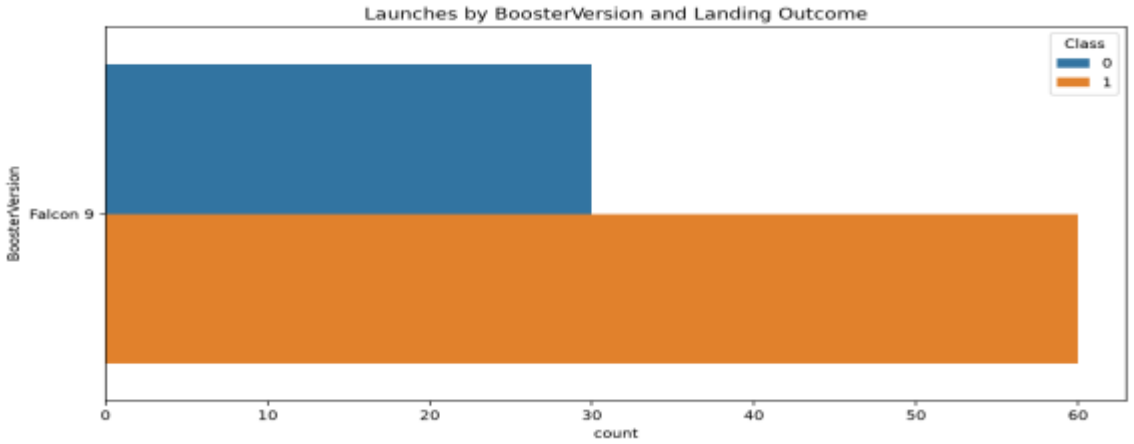
Payload mass distributions overlap across the two landing classes.

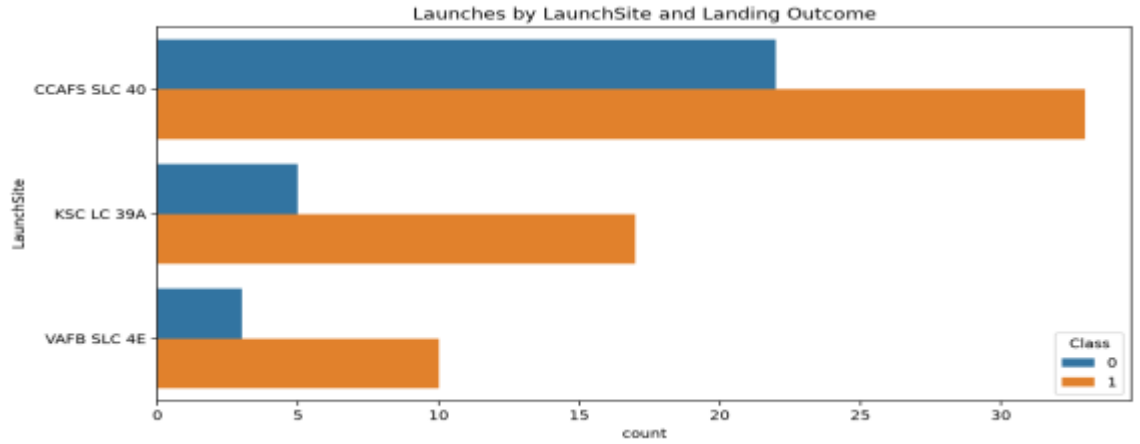


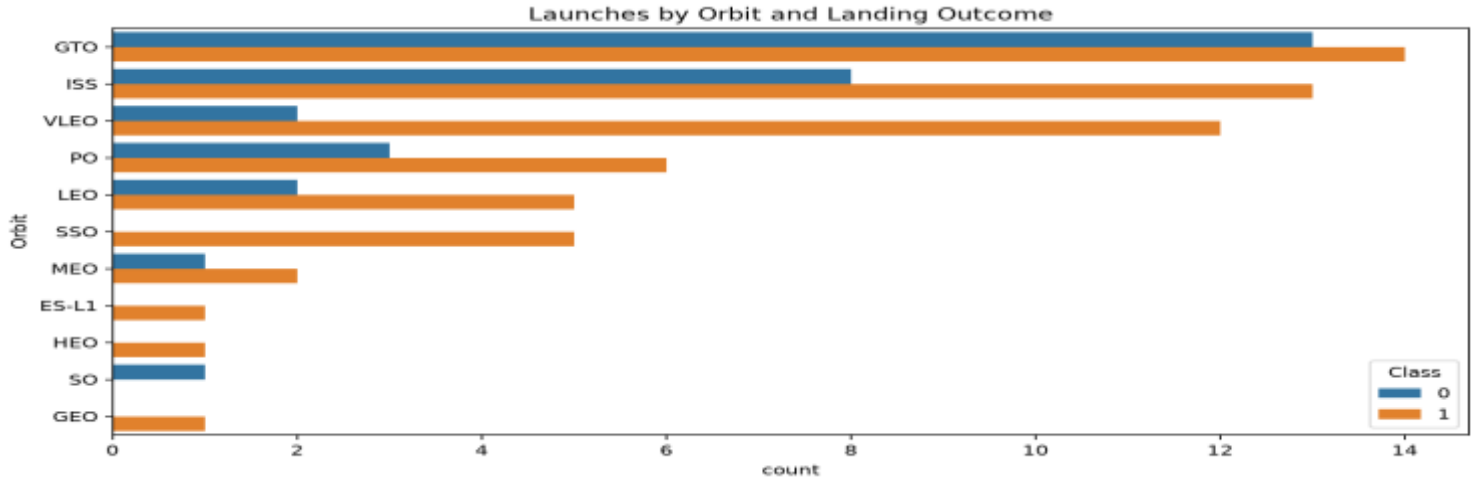
The plot suggests payload mass alone does not cleanly separate successful and unsuccessful landing outcomes.

Exploratory Data Analysis — Launch Characteristics

Launch counts vary by booster, launch site and orbit







SQL Analysis — Launch Site & Orbit

Launch volume by site: CCAFS SLC 40 = 55 launches; KSC LC 39A = 22; VAFB SLC 4E = 13.

Observed landing rates by site: KSC LC 39A = 77.3%; VAFB SLC 4E = 76.9%; CCAFS SLC 40 = 60.0%.

Orbit-level rates vary: SSO = 100% (5 records), GTO = 51.9% (27 records), ISS = 61.9% (21 records).

Small orbit groups (for example, HEO and GEO with one record each) should not be interpreted as reliable evidence of superior performance.

Machine Learning — Model Comparison

GridSearchCV (3-fold) tuned four classification algorithms

Model	CV F1	Test accuracy	Precision	Recall	Test F1
Decision Tree	0.882	0.889	0.917	0.917	0.917
Logistic Regression	0.864	0.833	0.800	1.000	0.889
KNN	0.889	0.833	0.800	1.000	0.889
SVM	0.889	0.778	0.750	1.000	0.857

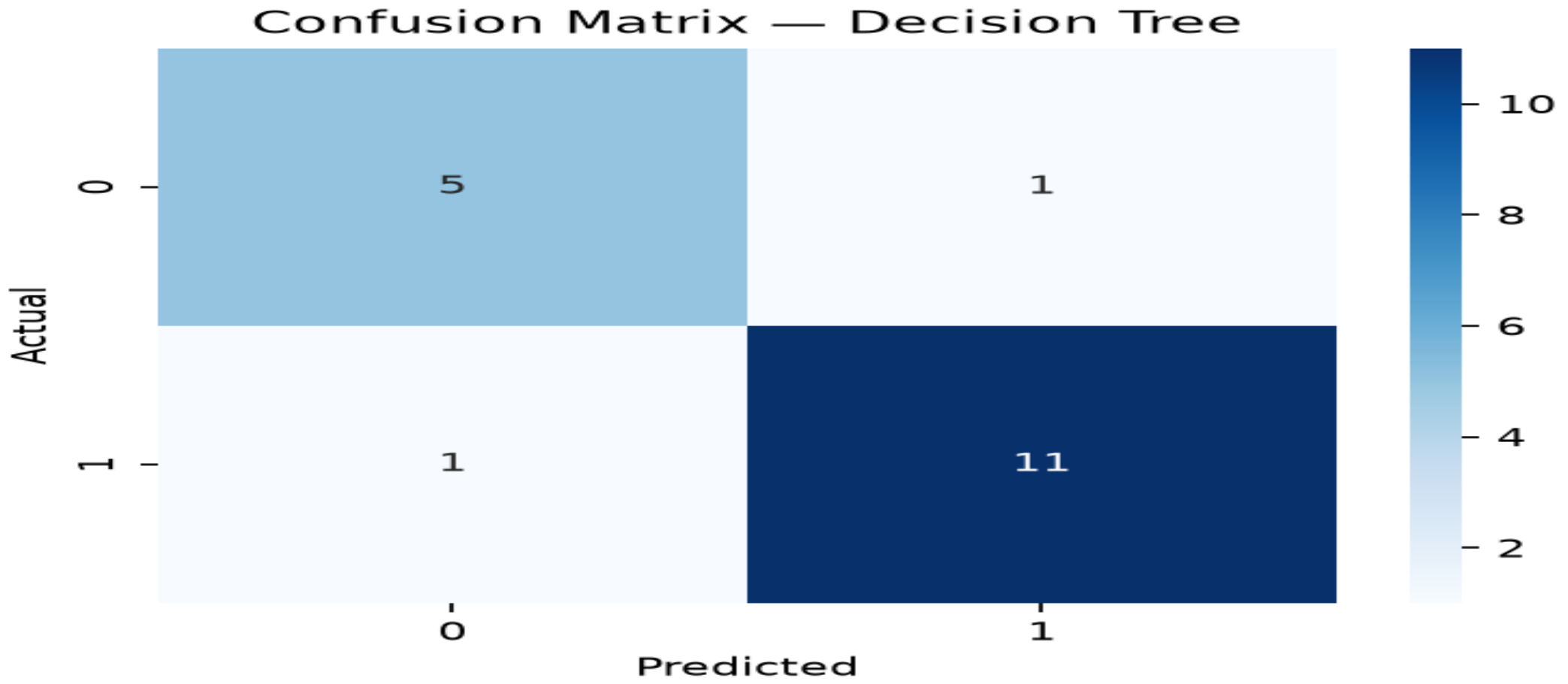
Decision Tree achieved the highest held-out test F1-score (0.917) and test accuracy (0.889).

Best Decision Tree settings: max_depth = 2; min_samples_split = 2.

KNN and Logistic Regression both achieved test F1 = 0.889; SVM achieved test F1 = 0.857.

Best Model — Decision Tree Confusion Matrix

Evaluation on the 18-record held-out test set.



TN = 5, FP = 1, FN = 1, TP = 11. The model correctly classified 16 of 18 test examples.

Conclusion & Limitations

The Decision Tree was the best model on this run, with 88.9% test accuracy and 0.917 F1-score.

The confusion matrix shows one false positive and one false negative; class-wise F1 was 0.83 for Class 0 and 0.92 for Class 1.

Launch-site and orbit summaries reveal differences in observed outcomes, but group sizes are uneven and some orbit categories have very few records.

Key limitation: only 90 labelled records and 18 test examples were used. Results from one train/test split can vary and are not sufficient to establish production-level performance.

Future work: evaluate with repeated stratified cross-validation, collect more labelled launches, check feature leakage and compare performance on a temporally separated test set.

Interactive Outputs & Reproducibility

Folium map: launch_sites_map.html — interactive markers with launch counts and observed landing rates.

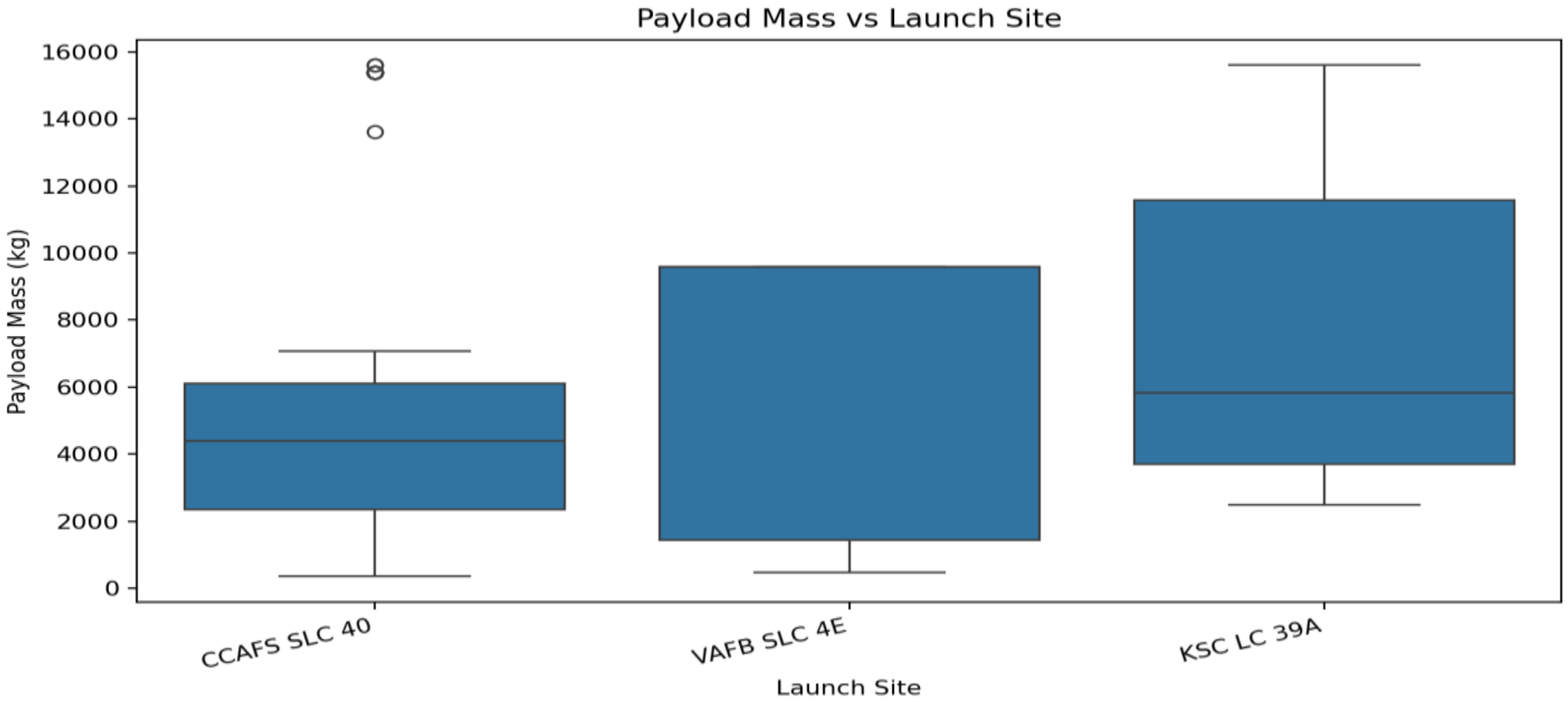
Plotly dashboard: interactive_dashboard.html — interactive launch and outcome visualizations.

Reproducibility: notebook 00_RUN_EVERYTHING.ipynb calls run_all.py to execute the pipeline and generate outputs.

Data source: IBM Skills Network prepared course datasets. This project uses prepared datasets rather than a fresh live SpaceX API extraction.

Additional EDA — Payload Mass vs Launch Site

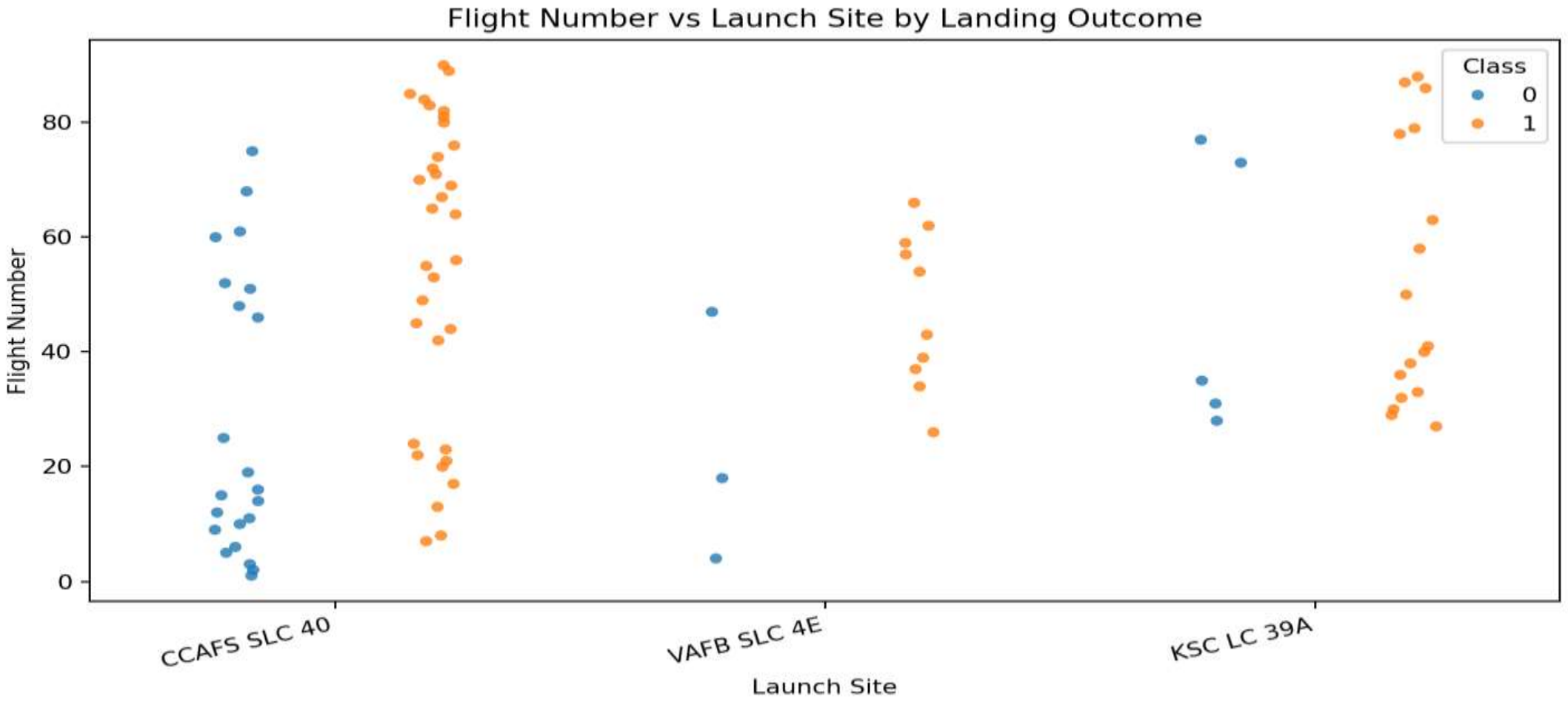
Compares payload-mass distributions across launch locations.



Interpretation: payload mass varies by launch site; distributions overlap and should not be treated as causal evidence.

Additional EDA — Flight Number vs Launch Site

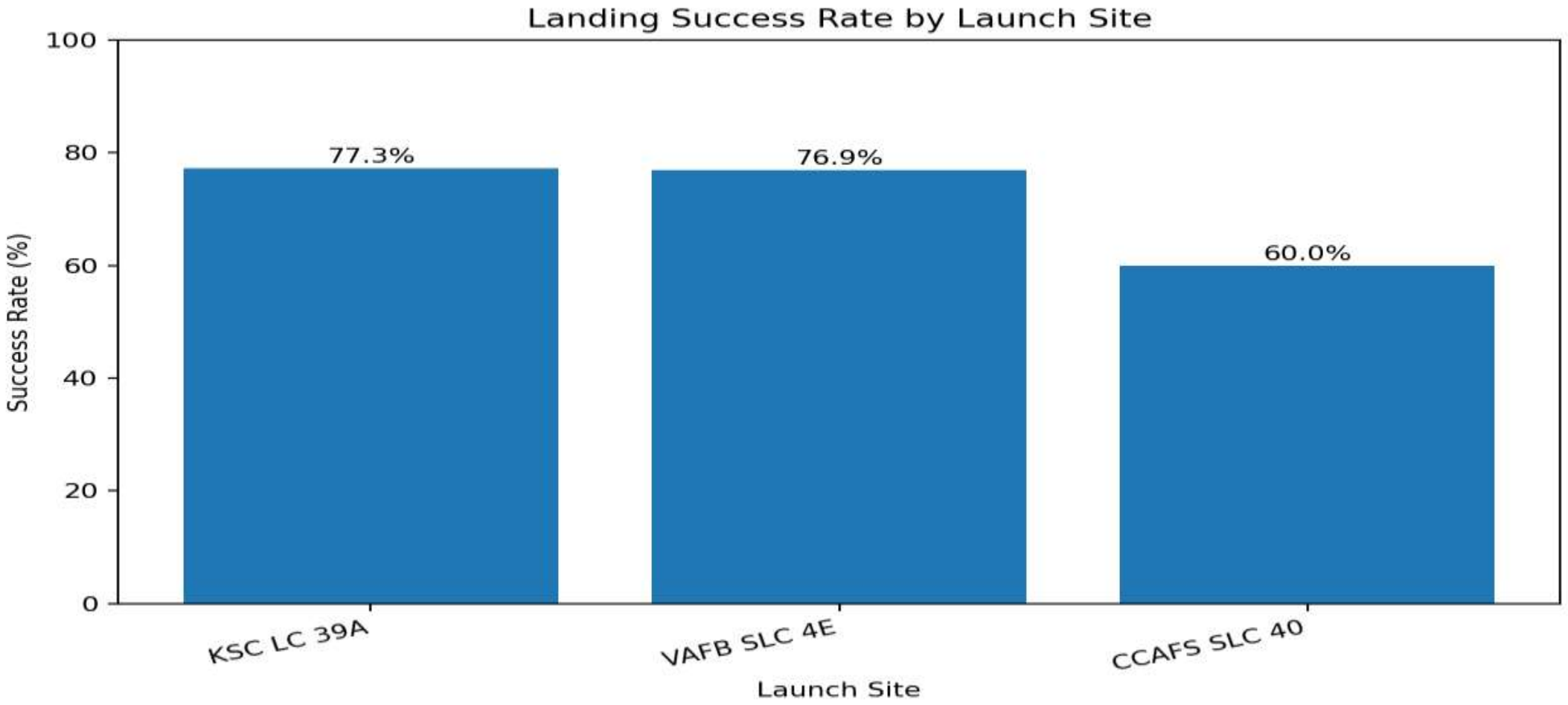
Shows flight sequence at each launch site, distinguished by landing class.



Interpretation: later flight numbers generally represent more recent launch experience; this pattern alone does not prove causation.

Landing Success Rate by Launch Site

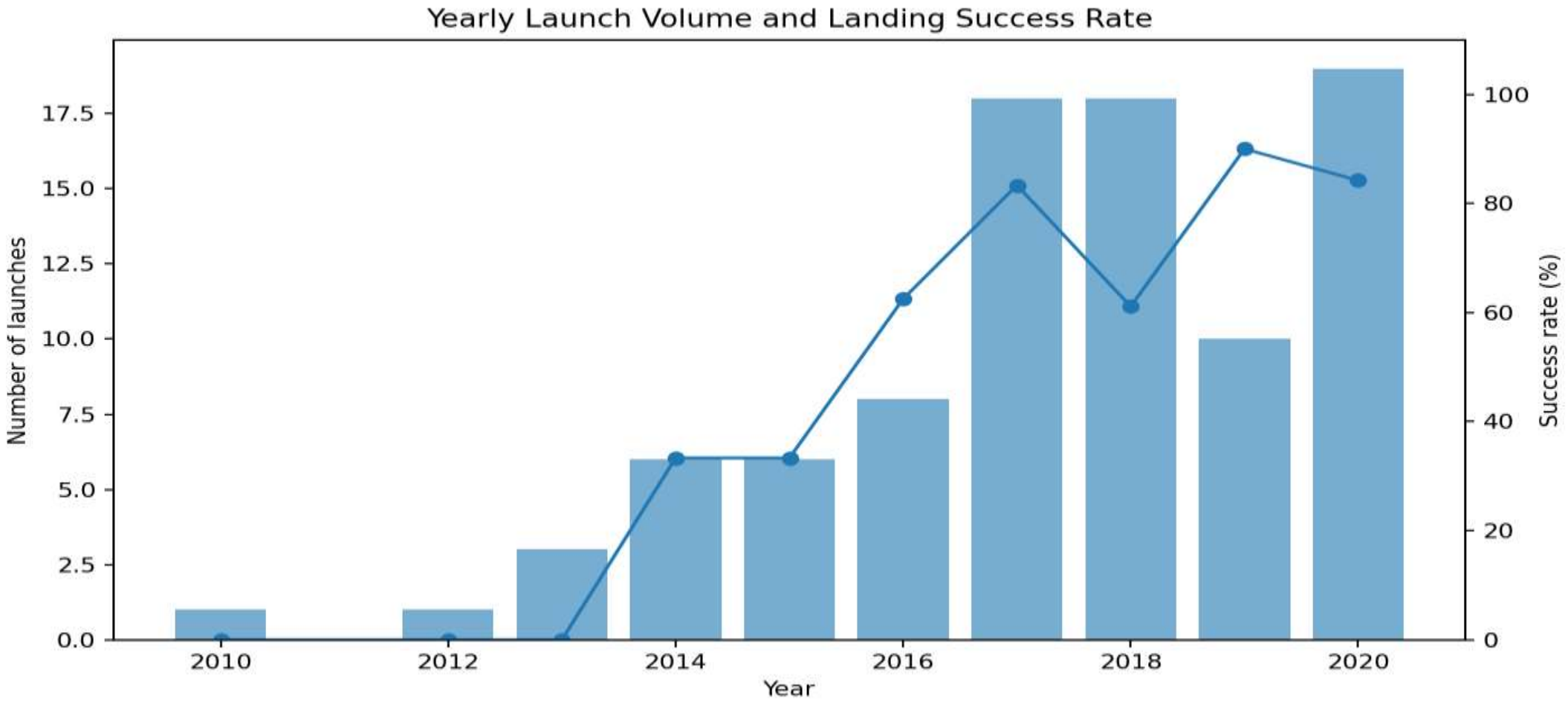
Observed Class 1 proportion for each launch site in the 90-record dataset.



KSC LC 39A: 77.3%; VAFB SLC 4E: 76.9%; CCAFS SLC 40: 60.0%.

Yearly Launch Volume and Landing Success Trend

Launch counts and observed landing success rate by year.



The yearly rates fluctuate; early years have very few records, so interpret percentages cautiously.

Expanded SQL Analysis

Additional queries calculated from the cleaned 90-record launch table.

- Unique launch sites: CCAFS SLC 40, KSC LC 39A and VAFB SLC 4E.
- Total payload mass: 549,446.35 kg; average payload mass: 6,104.96 kg.
- Success/failure by site: CCAFS SLC 40 — 33 successful, 22 unsuccessful.
- KSC LC 39A — 17 successful, 5 unsuccessful.
- VAFB SLC 4E — 10 successful, 3 unsuccessful.
- Yearly trend: 2020 had 19 launches with an 84.21% observed success rate; 2019 had 10 launches with 90.00%.
- These are descriptive results from the prepared dataset, not causal estimates.